

AYMEN MABROUK

TECHNICIAN IN
EMBEDDED SYSTEMS
AND INTERNET OF
THINGS

PROFILE

Computer Engineer specialized in IoT and Embedded Systems. Passionate about technology with a strong focus on Flutter development for mobile applications. Experienced in integrating real-time messaging and IoT solutions, with a commitment to delivering innovative, user-centric projects. Proficient in modern programming practices and agile methodologies, particularly Scrum. Adept at creating optimized, impactful applications by leveraging cutting-edge technologies and a creative problem-solving approach.

PROJECTS

2023-2024

1. Call Center Application

Description: Developed a mobile application for a call center, allowing customers to select the agent they wish to speak with.

Technologies Used: Flutter, MongoDB

Key Features:

- Intuitive user interface for easy navigation.
- Agent selection system based on preferences and interaction history.

2024

2. Medical Application for New Mothers

Description: Designed a mobile application to help new mothers track their baby's breastfeeding and vaccination schedules.

Technologies Used: Flutter, Firebase

Key Features:

- Automatic reminders for breastfeeding and vaccinations.
- Breastfeeding schedule tracking with personalized notifications.

+216 94-231-159

aymenmabrouk803@gmail.com

Tunisia, Monastir

<https://mabroukaymen1.github.io/portfolio.github.io/>

EDUCATION

2022 - 2025

HIGHER INSTITUTE OF COMPUTER SCIENCE - MAHDIA

- Bachelor's Degree in Embedded Systems and Internet of Things

2021 - 2022

KHNISS HIGH SCHOOL - KHNISS, MONASTIR

- Baccalauréat

SKILLS

- Nodejs
- Flutter
- MongoDB
- Postman
- java
- python
- C

PERSONAL SKILLS

- Creativity
- Communication
- Negotiation
- Critical Thinking
- Autonomy

LANGUAGES

- Arabic (Native)
- French (Intermediate)
- English (B1 - Intermediate Level)

3. Port Service Management System

Description: Developed a mobile app for managing port operations, equipment maintenance, and employee workflows. Features include real-time tracking, push notifications, offline support, and secure Firebase integration.

Technologies Used: Flutter, Firebase, REST API, Provider

Key Features:

- Role-based login and user management.
- Real-time tracking of port operations
- Employee attendance and task assignment

■ 2024 – 2025 | Final Year Project

4. Smart Office Automation Mobile App – Beemo Assistant

Description: Developed a smart automation mobile application integrated with the Beemo office assistant robot, enabling seamless office automation, robot control, and task execution. The app supports real-time interaction with smart devices and the robot using voice commands in English. Built for scalability and smooth user experience across platforms.

Technologies Used: Flutter · Firebase (Firestore, Realtime DB, Auth) · Vosk (English ASR) · Gemini AI · TCP/IP · REST API · WebSockets · Provider

Key Features:

- Real-time robot and device control with instant feedback and responsiveness.
- Voice command processing using the Vosk English ASR model for speech recognition and Gemini AI for intent analysis and command execution.
- TCP/IP-based communication with Beemo for navigation, expression control, and task execution.
- Secure QR code pairing for authenticated robot and device connections.
- Natural language interface with a chat-style UI for command entry, feedback, and task tracking.
- Firebase integration for data synchronization, user authentication, and real-time updates.